

Semantic-Motion Engine

Project Title: Semantic-Motion: Multi-Granularity Semantic Engine for Loco-Manipulation

Abstract

This project proposes **Semantic-Motion**, a multi-granularity semantic engine designed for loco-manipulation tasks. The goal is to establish an automated Vision-Language-Action (VLA) data loop that bridges high-level macro-intents and low-level micro-instructions, enabling scalable instruction synthesis, motion refinement, and multi-view rendering for embodied robotics.

Streams of Project

- **Perception:** Automated labeling of macro-intents and micro-instructions using Vision-Language Models (VLMs).
- **Augmentation:** LLM-driven text rewriting for diverse, high-fidelity instruction synthesis.
- **Refinement:** Filtering out low-quality actions and samples with low confidence scores.
- **Rendering:** Converting 3D motion into multi-view visual signals via physical engines.

Keywords

Loco-Manipulation, VLA Data Loop, Multi-view Synthesis, Automated Annotation

Mentor: Prof. Taku Komura

Group Members: Gai Yifan, Yang Tongzhou, Li Chenyang